

Comparative Risk of Hyperglycemia with PI3K, AKT, and mTOR Inhibitors in Breast Cancer

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FINAL PUBLICATION NUMBER (FPN): 530P



INTRODUCTION

- Targeting the PI3K-AKT-mTOR (PAM) signaling pathway has become a central therapeutic strategy in advanced breast cancer, particularly in hormone receptor-positive disease.
- Multiple agents targeting the PAM pathway are in clinical use or late-stage development, including the PI3K inhibitors alpelisib, inavolisib, and taselisib; the AKT inhibitor capivasertib; and the mTOR inhibitor everolimus.
- Although these agents enhance disease control, pathway inhibition can impair insulin signaling and commonly cause treatment-related hyperglycemia, sometimes requiring dose interruption, discontinuation, or metabolic management.

METHODS

- A literature search was performed across PubMed/MEDLINE, Embase (OVID), CENTRAL, ClinicalTrials.gov, and Web of Science from database inception through January 1, 2026.
- A total of 4,104 records were identified and screened
- Eligible studies were randomized controlled trials (RCTs) in breast cancer comparing a PAM pathway inhibitor at a standard dose versus a control regimen lacking that agent
- Standard doses included for the analysis were alpelisib 300 mg, inavolisib 9 mg, taselisib 4 mg, capivasertib 400 mg twice daily, and everolimus 10 mg.

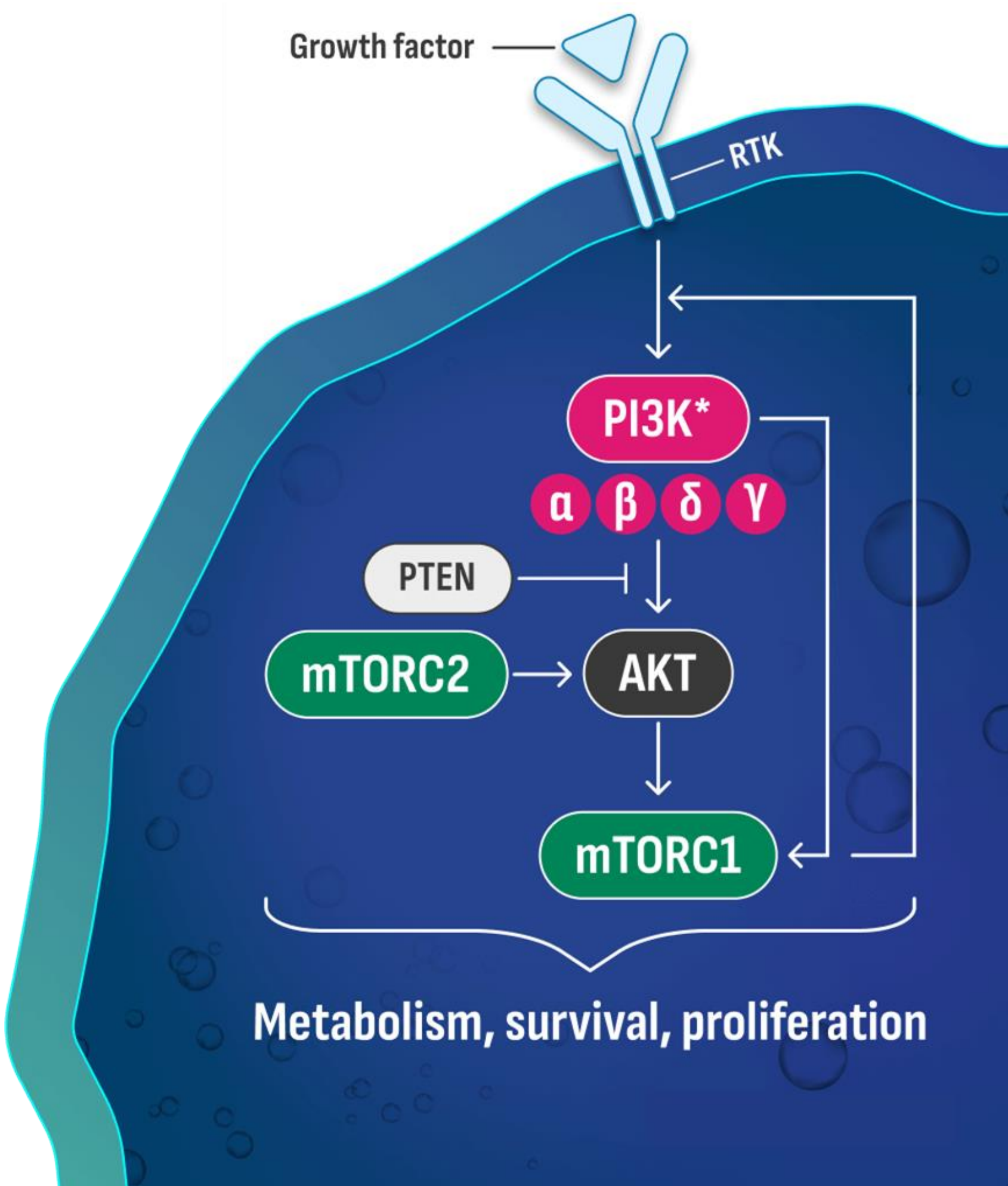
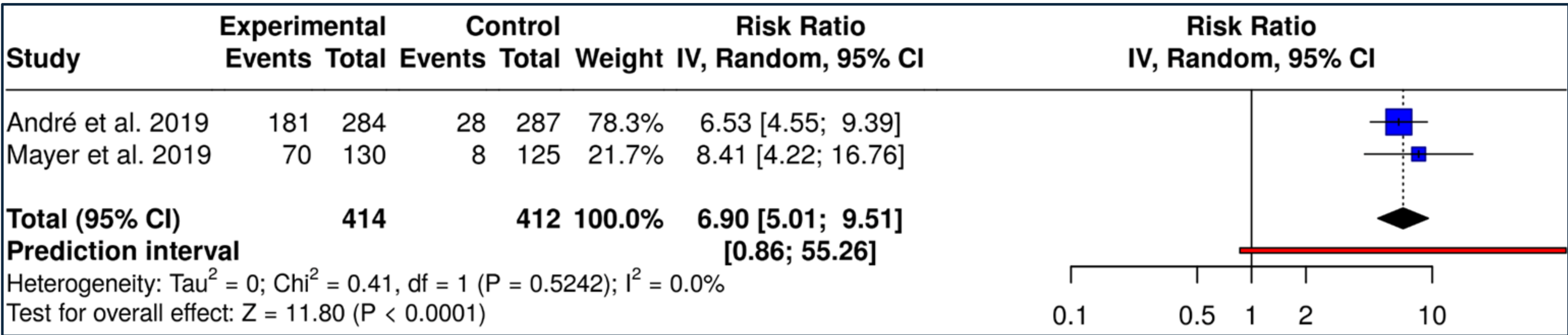
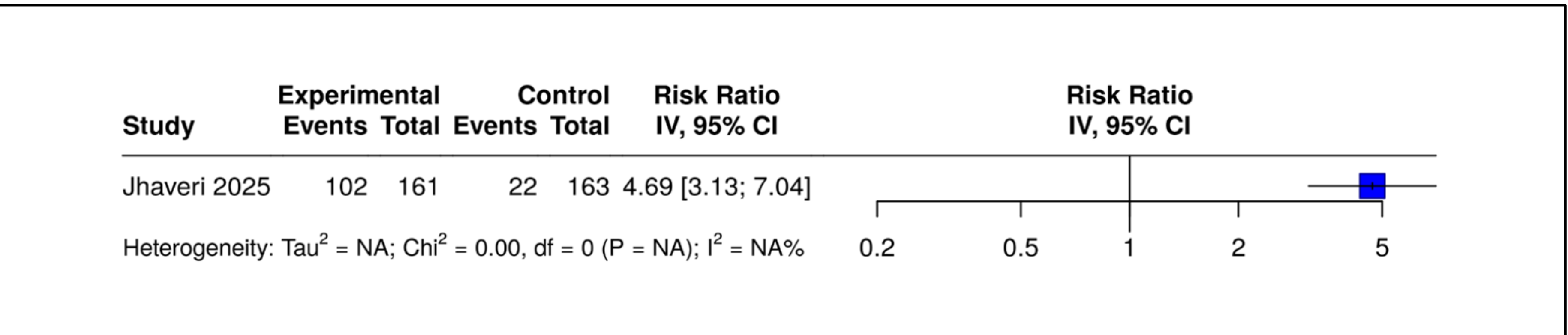


Figure 1 Overview of the PI3K/AKT/mTOR (PAM) Signaling Pathway

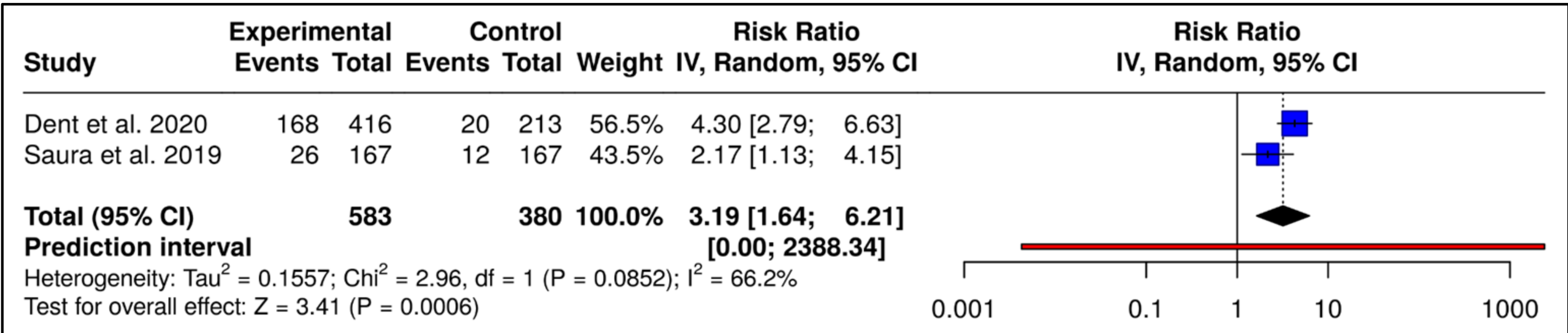
Alpelisib



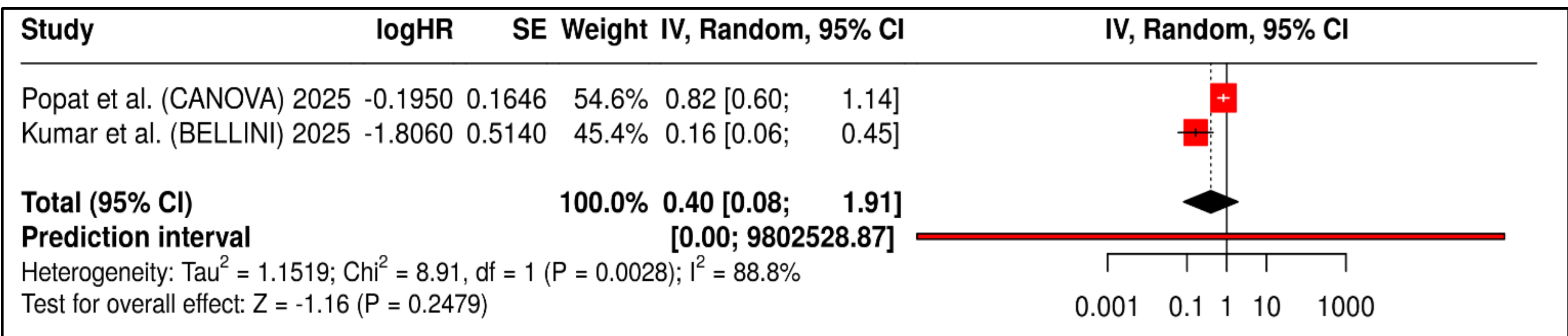
Inavolisib



Taselisib



Capivasertib



Everolimus

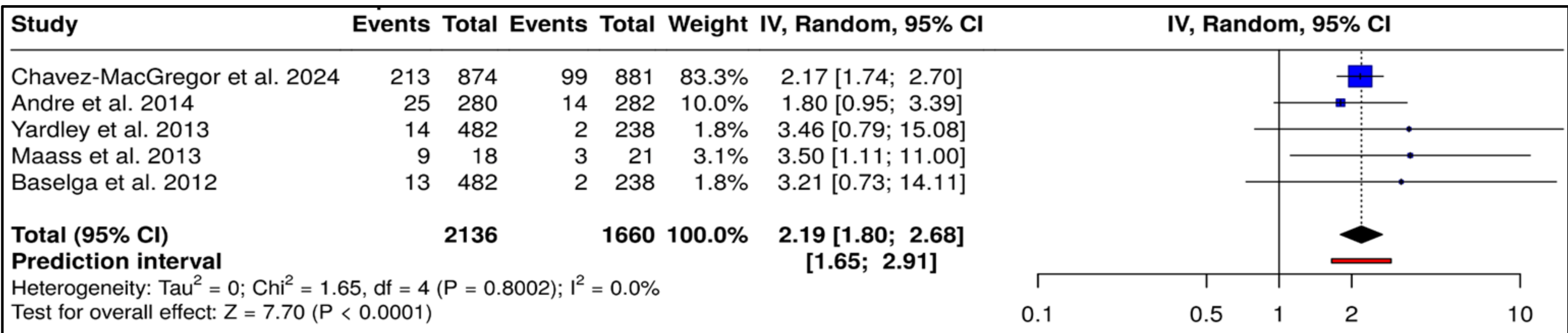


Figure 2 Summary of Pooled Outcomes for any-grade hyperglycemia

RESULTS

- Across 14 randomized trials (n = 6,940), all PAM pathway inhibitors increased hyperglycemia risk versus control regimens
- Alpelisib demonstrated the highest risk (any-grade RR ~6.9; grade ≥3 RR ~58.8)
- Inavolisib also showed a high risk signal (any-grade RR ~4.7; grade ≥3 RR ~23.3;)
- Capivasertib and taselisib showed intermediate risk (any-grade RR ~3.5 and ~3.2; grade ≥3 RR ~10.0 and ~14.6, respectively)
- Everolimus demonstrated the lowest risk (any-grade RR ~2.2), with no statistically significant increase in grade ≥3 hyperglycemia

CONCLUSION

- All PAM pathway inhibitors were associated with an increased risk of hyperglycemia compared with control regimens, with clear stratification by drug class.
- PI3K inhibitors demonstrated the highest relative risk for both any-grade and grade ≥3 hyperglycemia, AKT inhibitors showed an intermediate risk, and mTOR inhibitors the lowest risk.
- This severity-dependent gradient likely reflects increasing proximity of pathway inhibition to insulin receptor signaling, with PI3Kα inhibition most directly disrupting glucose homeostasis.
- Interpretation is limited by heterogeneity in trial design and limited randomized data for newer agents.

CLINICAL IMPORTANCE:

Hyperglycemia is a clinically significant, mechanism-based (on-target) toxicity of PAM pathway inhibitors with potential for severe manifestations

Risk varies markedly by drug class: PI3K inhibitors carry the highest risk, followed by AKT inhibitors, with lower risk from mTOR inhibitors.

Baseline A1c and early glucose monitoring are critical when these agents are initiated. Active glucose management is often required, including initiation or escalation of oral antihyperglycemic agents and/or insulin therapy

Severe (grade ≥3) hyperglycemia is not rare and can require insulin, hospitalization, or treatment interruption.